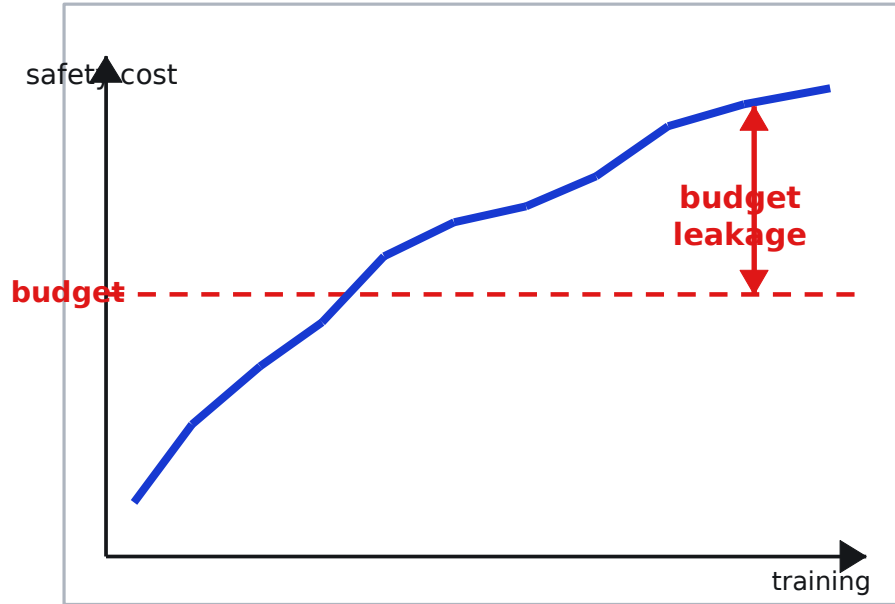


Penalty-based safe RL

scalarized objective

$$\max_{\pi} J_r(\pi) - \lambda^\top \mathbf{J}_c(\pi)$$

One penalty vector must price every safety channel.



Typical failure mode

A fixed or slowly adapted penalty can trade away a rare channel while improving the scalarized objective.

SafeTransport

one capacitated γ -flow bridge

1. Encode K constraints



Bridge impact

c_1 with budget b_1
induces capacity $S^{(1)}$

hard



Liquid spill

c_2 with budget b_2
induces capacity $S^{(2)}$

soft



Contact force

c_3 with budget b_3
induces capacity $S^{(3)}$

hard



Fire / tipping

c_4 with budget b_4
induces capacity $S^{(4)}$

hard

Heterogeneous semantics stay separate until composition.

2. Compose one capacity

Hard minimum

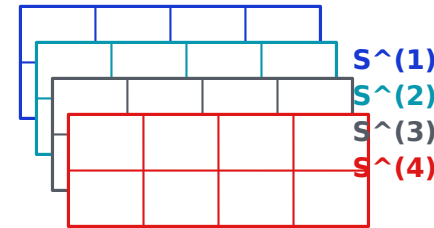
$$S_{ss'}^{(k)} = \frac{b_k \omega_{ss'}^{(k)}}{c_k(s)}$$

$$S = \min_{k \in [K]} S^{(k)}$$

theorem-level simultaneous-channel certificate

Smooth Compose(T_c)

calibrated alternative; no hard-min certificate



one solver matrix S

epsilon selects a feasible-flow optimum at fixed S

3. Solve once

capacitated γ -flow Sinkhorn

$$F_\varepsilon^* = \arg \min_{F \in \mathcal{F}_S}$$

$$\langle D, F \rangle - \varepsilon H(F)$$

Gibbs scaling + capacity truncation

capacity-feasible target flow

0.45	0.45	0.21
0.65	0.55	0.02
0.12	0.34	0.34

$$F_\varepsilon^* \leq S$$

FLOW TARGET

F^* certified

DEPLOYED POLICY

PPO follows F^* via clipped weights

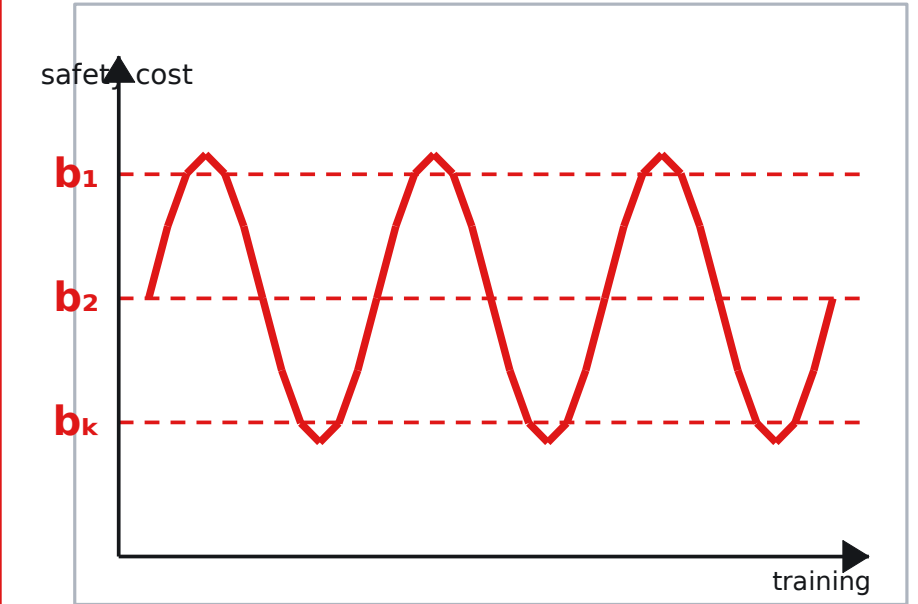
certificate checks deployed-flow gap

Constraint-based safe RL

explicit CMDP budgets

$$\max_{\pi} J_r(\pi) \quad \text{s.t.} \quad J_{c_k}(\pi) \leq b_k, \quad k \in [K]$$

Budgets are explicit, but outer controllers or local projections must coordinate active constraints.



Typical failure mode

Coupled dual or projection updates can oscillate near multiple active budgets and become brittle.

Conceptual positioning: penalties → one SafeTransport capacity solve → projections

PPO / SAC

PPO-Lag /
FOCOPS

SafeTransport

one inner capacity projection

CPO

CUP / hard projection

unconstrained or penalty

external safety controller

local constrained step

constraint-dominant

Inside SafeTransport at fixed S : $\varepsilon \rightarrow 0+$ gives the hard-capacity LP optimum; $\varepsilon \rightarrow \infty$ gives the maximum-entropy feasible flow. Capacity $F \leq S$ remains active.